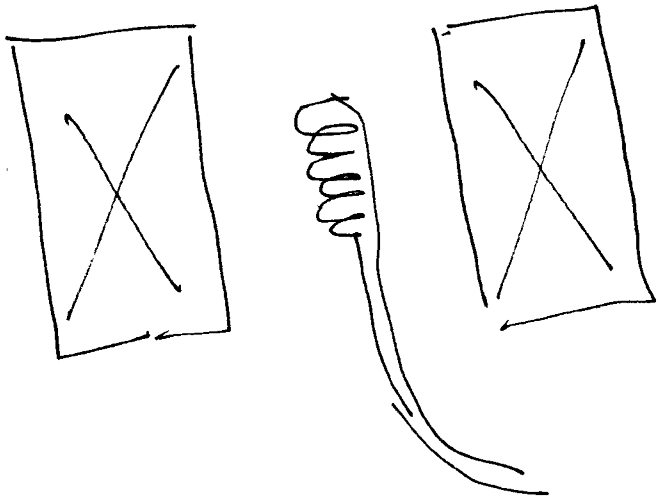
Basic Idea



→ A long solenoid generates a time dependent magnetic field, $H(t) = H_0 \cos \omega t$

$$\omega = (2\pi * 50)\ \frac{\text{rad}}{\text{sec}}$$

→ A small pick up coil measures the induced voltage. The sample is placed inside the pick up coil.

Aim: We want to plot $\underline{\underline{M \text{ vs } H}}$ for the sample.

→ What is measured is H vs t ← channel 1

M vs t ← channel 2

→ Go to "x y mode" to eliminate time & directly plot M vs H.

Magnetization is measured using Lenz's law:

Induced voltage (e_2)

$$e_2 = -\frac{\partial \Phi_B}{\partial t} = -\frac{\partial}{\partial t}\left[A_c \mu_0 H(t) + A_s \mu_0 M(t)\right]$$

Signal on the y-axis (channel 2)

→ This should be the time dependent magnetization of the sample, following the applied magnetic field.

Signal on the x-axis (channel 1)

→ The magnetic field seen by the sample

= (magnetic field in the solenoid) − demagnetizat

→ Channel 2

Pick up voltage − the signal from empty coil.

(a) we need to modify e_2 such that the $H(t)$ part of the e_2 signal is completely eliminated & we only have a signal proportional to $M(t)$ (2)

So (1) Integrate $e_2 \longrightarrow e_3 = -\int e_2 \, dt$

(2) phase shift e_3 such that the signal is in phase with the magnetic field.

[Since $H = H_0 \cos \omega t$

$$\frac{\partial H}{\partial t} \propto \frac{H_0}{\omega} \cos(\omega t + \phi);$$

$\rightarrow$ The phase knob brings ϕ to zero

(3) ~~Add some~~ Subtract a fraction of the field signal, ~~$H_0 \cos$~~ so that only the magnetization part survives ~~in the~~ in e_3.

(4)

$$\boxed{H = H_a - NM}$$; N: demagnetization factor

H: the field the sample sees is less than what is applied by a constant factor N, called the demagnetization factor.

$H = H_0 \cos \omega t$; $\omega = (2\pi * 50)$ rad/sec

There is a circuit that generates a voltage e_1 proportional to the applied field H_a.

$$\boxed{e_1 = C_1 H_a}$$

$$B = \mu_0 H + J$$

$$\Phi_B = A_c \mu_0 H + A_s J$$

The induced voltage

$$e_2 = -\frac{\partial \Phi_B}{\partial t}$$

This signal is integrated & phase shifted with an operational amplifier of gain g_1

$$e_3 = -g_1 \int e_2 \, dt$$

$$e_3 = \frac{g_1 A_c H \mu_0}{\omega} + \frac{g_1 A_s J}{\omega}$$ [Also a phase shift: H & J are periodic signals

Now, there is a differential amplifier with gain g_y

$$e_y = -g_y(e_1 - e_3)$$

$$= -g_y C_1 H_a + e_3 g_y$$

$$e_y = -g_y C_1 H_a + g_y \frac{g_1 A_c H}{w} \mu_0 + g_y \frac{g_1 A_s}{w} J$$

Write H as H_a; $H = H_a - NM$

in the second term.

$\Rightarrow$

$$e_y = -g_y C_1 H_a + \frac{g_y g_1 A_c}{w} \mu_0 H_a$$

$$- \frac{g_y g_1 A_c}{w} \mu_0 NM + g_y \frac{g_1 A_s}{w} J$$

$\mu_0 M = J$

Now adjust C_1 such that the pick up signal cancels. [Note that this will change the x-axis scale also].

i.e. choose C_1 such that

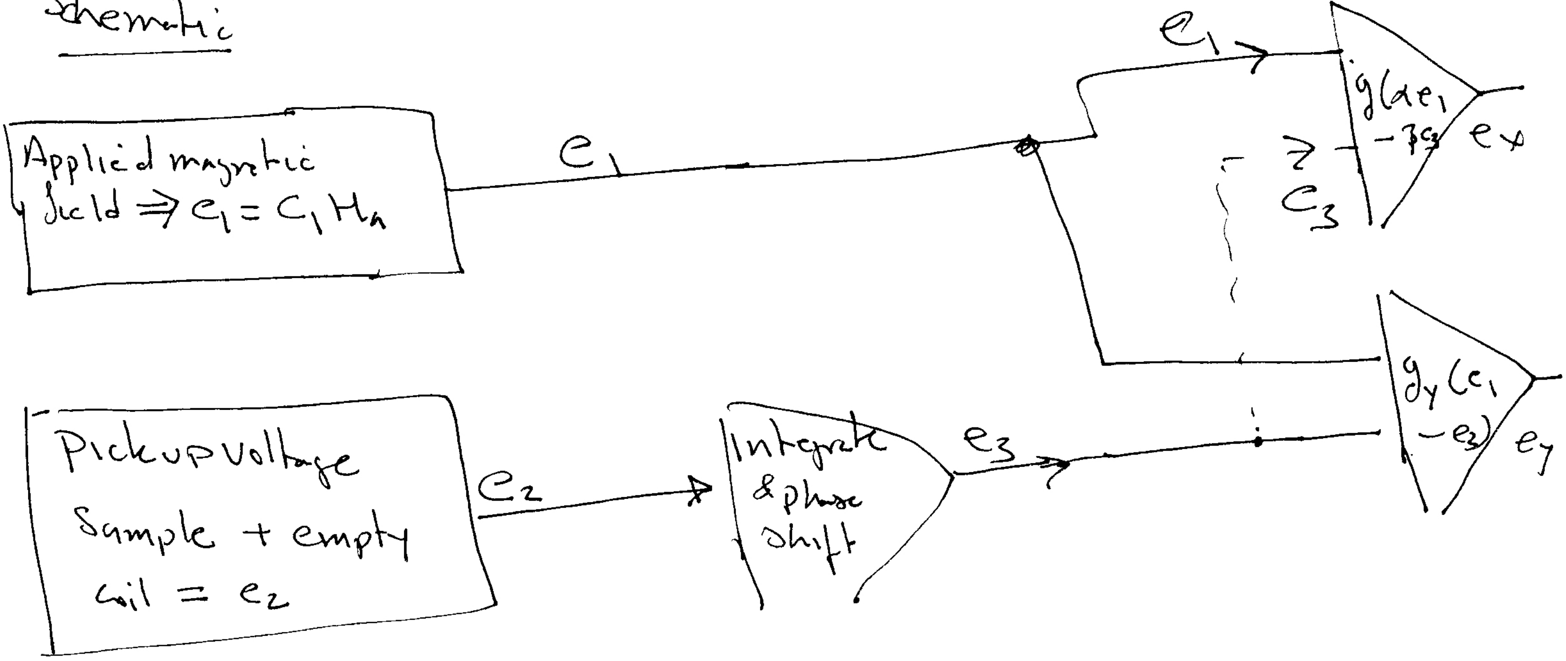
$$g_y C_1 = g_y \frac{g_1 A_c}{w} \mu_0 \Rightarrow \boxed{C_1 = \frac{g_1 A_c \mu_0}{w}}$$

After appropriately choosing C_1,

$$e_y = -\frac{N M \mu_0 g_y g_1 A_c}{\omega} + \frac{g_y g_1}{\omega} A_s \mu_0 M$$

$$e_y = \frac{g_y g_1}{\omega} \mu_0 M \left(A_s - A_c N\right)$$

Schematic



Ch1 ⇒ e_x : measures the magnetic field applied by the big solenoid. This field has to be corrected for the demagnetization ⇒ subtract part of the magnetic response from applied field.

$$H = H_a - NM$$

Ch2 ⇒ e_y : Measure sample magnetization via a pick up coil. The ~~sample~~ signal has to be corrected for the pick response of empty coil A

The x-axis [channel 1 signal].

We already had

$$e_1 = C_1 H_a$$

with C_1 already chosen ~~to be~~

$$\boxed{C_1 = \frac{g_1 A_c \mu_0}{w}}$$

← This value ensured that the channel 2 signal is only from the sample & the empty coil contribution is subtracted out.

To depict the hysteresis loop in real time, the x axis should be the field seen by the sample & not the applied field, that is we need to correct for the demagnetization factor.

$H = H_a - NM$; we want $M(H)$ & not $M(H_a)$.

N: demagnetization factor.

⇒ take part of the magnetization signal & ~~feed it to the~~ ~~add~~ subtract it ~~to the~~ from the signal e_1.

⇒

$$e_x = g_x(\alpha e_1 - \beta e_3)$$

$\Leftrightarrow$

$$e_x = g_x(\alpha e_1 - \beta e_3)$$

$$= g_x \alpha \frac{g_1 A_c}{\omega} H_a \mu_0 - g_x \beta\left(\frac{A_c \mu_0 H g_1}{\omega} + \frac{A_s \mu_0 M g_1}{\omega}\right)$$

$$e_x = \frac{g_x g_1}{\omega} \mu_0 A_c \left[\alpha H_a - \beta H - \beta M \frac{A_s}{A_c}\right]$$

now $H = H_a - NM$, eliminate H_a

$$e_x = K\left[\alpha H + \alpha NM - \beta H - \frac{\beta A_s}{A_c} M\right]$$

with $K = \frac{g_x g_1}{\omega} \mu_0 A_c$

or

$$e_x = K\left[(\alpha - \beta) H + \left(\alpha N - \frac{\beta A_s}{A_c}\right) M\right]$$

Now if we tweak α & β such that

$\alpha = \frac{A_s}{A_c}$ & $\beta = N$, the second term becomes zero

&

$$e_x = K\left[(\alpha - \beta) H\right]$$

or

$$H = \frac{e_x}{K(\alpha - \beta)} = \frac{e_x}{K\left(\frac{A_s}{A_c} - N\right)} =$$